

PCB DESIGN

COMPREHENSIVE CHEAT SHEET

Schematic Design · PCB Layout · Design Workflow · KiCAD Tricks & Techniques · Signal Integrity · EMC · Manufacturing

Professional Engineering Reference | 26 Comprehensive Sections

1. PCB DESIGN WORKFLOW — FULL PROCESS

COMPLETE PCB DESIGN FLOW (SCHEMATIC → ASSEMBLED BOARD)			
PHASE 1 — REQUIREMENTS & PLANNING			
Define requirements	Voltage, current, signal freq, temp range, size, cost	Standards	IPC-2221 (generic), IPC-2222 (rigid), IPC-6012 (qual)
Layer count decision	2-layer (simple), 4-layer (mixed), 6+ (high-speed/RF)	Board size	Verify with mechanical team before starting layout
Component selection	Availability, footprint, package thermal, cost, lead time	BOM planning	Check JLCPCB/LCSC/Mouser stock early in design
PHASE 2 — SCHEMATIC DESIGN			
Create symbols	Draw or import — match datasheet pin names exactly	Annotate	R1,R2... C1,C2... U1,U2... — consistent prefixes
Hierarchical design	Top-level sheet calls sub-sheets per functional block	Flat design	Single sheet — OK for simple boards (<50 parts)
ERC (Electrical RC)	Run before layout — fix all errors, review warnings	Net naming	Name critical nets: VCC_3V3, GND_AGND, CLK_48MHz
BOM generation	Export CSV with MPN, value, footprint, quantity	Cross-check	Verify every component has a valid footprint assigned
PHASE 3 — PCB LAYOUT			
Import netlist	Sync schematic → PCB (KiCAD: Update PCB from Schematic)	Board outline	Set board edge (User.Cuts or Edge.Cuts layer)
Place components	Group by function, power first, then signal	Constraint zones	Define keepout, copper pour, via restrictions
Route power	Wide traces first: VCC, GND, PWR rails	Route signal	Short, direct, avoid acute angles (use 45° or curves)
Pour GND plane	Fill remaining copper with GND on all layers	Check DRC	Fix all errors before sending to fab
PHASE 4 — VERIFICATION & OUTPUT			
DRC	Design Rule Check — clearance, short, unconnected	3D check	Verify 3D model for mechanical fit + collision
Gerber export	One file per layer + drill file + drill map	Fab review	Use fab's DFM checker (JLC, PCBWay, OSHPark)
PCBA output	Centroid (pick & place) + BOM for assembly	Silkscreen	Verify all ref designators visible, not under parts
PHASE 5 — PROTOTYPE & VALIDATION			
Bare board test	Continuity, shorts, impedance if controlled	Assembly	Reflow (SMD) / wave (THT) / hand solder
Bring-up sequence	Power rails first → core → peripherals → firmware	Oscilloscope	Check clock, power ripple, signal integrity
ECO (Eng Change)	Document every change: what, why, who, when	Revision	Rev A (proto), Rev B (fix), Rev C (production)
NOTE: Follow IPC-2221B for generic design requirements. Every design decision should be traceable to a requirement. Never skip ERC/DRC — fix all errors before proceeding.			

2. SCHEMATIC DESIGN FUNDAMENTALS

SCHEMATIC DRAWING STANDARDS & CONVENTIONS			
ORGANISATION & READABILITY			
Signal flow	Left to right, top to bottom — inputs left, outputs right	Sheet border	Include title block: project, rev, date, author, sheet

Net labels vs wires	Use net labels for signals crossing far distances	Power symbols	Use VCC/GND power symbols — not wired to rails
Off-sheet refs	Hierarchical pins or global labels for multi-sheet	Net ties	Connect different GND domains (AGND/DGND) with net tie
No connect (X)	Mark unused pins explicitly — prevents ERC errors	Dangling wire	Never leave wire ends unconnected — always terminate
Component spacing	Leave space around ICs for annotations and readability	Wire angles	Horizontal/vertical only — no diagonal wires
REFERENCE DESIGNATORS (IPC-2612)			
R — Resistor	R1, R2, R3...	C — Capacitor	C1, C2, C3...
L — Inductor	L1, L2...	U — IC/Module	U1, U2... (integrated circuit)
Q — Transistor	Q1, Q2... (BJT/FET/IGBT)	D — Diode	D1, D2... (diodes, LEDs, Zener)
J — Connector	J1, J2... (jacks, headers, terminals)	SW — Switch	SW1, SW2...
F — Fuse	F1, F2...	TP — Test Point	TP1, TP2... (probe points)
FB — Ferrite Bead	FB1, FB2...	Y — Crystal	Y1, Y2... (oscillators, resonators)
POWER & GROUND HIERARCHY			
VBAT	Battery voltage (unregulated, high impedance source)	VIN / VMAIN	Primary input supply (before regulator)
VCC / VBUS	Main regulated supply (e.g. 5V, 3.3V)	VCORE	Digital core supply (1.0-1.8V for MCU/FPGA)
VDDA / AVDD	Analogue supply — separate from digital VCC	VREF	Precision reference for ADC/DAC
AGND	Analogue ground — connect to DGND at single star point	PGND	Power ground — high-current return paths
DGND / GND	Digital ground — main logic return	Shield GND	Chassis/screen ground — connect at one point
DECOUPLING CAPACITORS IN SCHEMATIC			
100nF ceramic	Place on every VCC pin of every IC (mandatory)	10µF bulk	Per power domain: near voltage regulator output
1µF ceramic	For ICs with multiple VCC pins (bridge 100nF & 10µF)	VCORE decoup.	100nF + 1µF per VCORE pin on MCU/FPGA
Value annotation	Always label value + voltage rating: 100nF/10V X7R	Symbol	Use generic cap symbol — assign value in properties
SCHEMATIC ANNOTATION & BOM DISCIPLINE			
MPN field	Manufacturer Part Number in component properties — e.g. MPN = RC0402FR-0710KL (Yageo 10k 1% 0402)		
Value format	Resistors: 10k, 4.7k, 100R Capacitors: 100nF, 10µF Inductors: 10µH Always include tolerance		
Footprint	Assign EARLY: Resistor_SMD:R_0402 Capacitor_SMD:C_0603 Package_SO:SOIC-8_3.9x4.9mm_P1.27mm		
NOTE: Golden rule: schematic is the engineering intent — it must be correct, readable, and complete BEFORE layout begins. A bad schematic creates a bad PCB.			

3. COMPONENT FOOTPRINTS & LAND PATTERNS

SMD PACKAGE SIZES & DIMENSIONS			
RESISTORS & CAPACITORS — IMPERIAL / METRIC			
0201 (0603 metric)	0.6×0.3mm — reflow only, no hand solder	0402 (1005)	1.0×0.5mm — small, standard for modern designs
0603 (1608)	1.6×0.8mm — hand-solderable, beginner-friendly	0805 (2012)	2.0×1.25mm — easy to solder, good for high voltage
1206 (3216)	3.2×1.6mm — high power, high voltage caps	1210 (3225)	3.2×2.5mm — large capacitors (10µF+)
2512 (6432)	6.4×3.2mm — power resistors (1W+)	Minimum pitch	0402 for production; 0201 for space-critical
IC PACKAGES			
SOT-23-3	3-pin, 0.95mm pitch — BJT, small regulators	SOT-23-5/6	5/6-pin, 0.95mm — dual gates, small ICs
SOIC-8	1.27mm pitch, 3.9 or 5.3mm wide — op-amps, drivers	SOIC-16	1.27mm pitch, 16 pins — logic, comms ICs
TSSOP-20/24	0.65mm pitch — dense logic, DACs, drivers	TQFP-32/48/64	0.8mm pitch — MCUs, CPLDs
LQFP-100/144	0.5mm pitch — STM32, larger MCUs	QFN-20..64	No-lead bottom pad — good thermal, needs reflow
BGA packages	0.5/0.8/1.0mm pitch — FPGAs, processors, DDR	BGA via-in-pad	Requires ECIA pad plating for reliability
THROUGH-HOLE (THT) PACKAGES			
DIP-8/14/16/28/40	2.54mm (100mil) pitch — ICs, legacy	TO-220	Power transistors, regulators — bolt to heatsink
TO-92	Plastic transistor — BC547, 2N2222	TO-247	High-power FETs/IGBTs > 10A
Axial R/C	1.27mm min lead pitch for 100mil grid	Radial C	Electrolytic — 2.54mm or 5.08mm pitch
LAND PATTERN DESIGN RULES (IPC-7351)			
Courtyard	Minimum clearance envelope around part (0.25mm min)	Fab layer	Shows actual component body outline
Courtyard clearance	0.25mm (IPC-7351B nominal), 0.5mm for hand assembly	Silkscreen	Do NOT overlap pads — move to fab layer if needed
Paste aperture	Paste area = 88% of pad area (default), adjust for QFN	Center pad	QFN/DFN thermal pad: cross-hatch paste, 50–70% coverage
Pad extensions	Extend pads beyond body for inspection / hand soldering	SOD-123 heel	Add 0.3mm heel extension beyond body line
CONNECTOR FOOTPRINTS			
2.54mm header	Standard Arduino-style, hand-solderable	1.27mm header	Compact, surface mount available
JST-PH 2.0mm	Common battery/sensor connector	JST-XH 2.5mm	Standard in many modules
USB Type-C	Use official footprint — check pad polarity carefully	RJ-45	Ethernet — usually shielded, check keepout
NOTE: Always verify footprints against actual component datasheet dimensions. Manufacturer footprints are NOT always correct in KiCAD libraries — measure a physical part or use IPC-7351 calculator.			

4. LAYER STACKUP & CONTROLLED IMPEDANCE

COMMON LAYER STACKUPS

2-LAYER STACKUP (1.6mm total)			
Layer 1 (Top)	Signal + Components + Power traces	Layer 2 (Bot)	GND plane (pour) + secondary signals
Core thickness	1.6mm FR4, $\epsilon_r = 4.3-4.7$	Copper weight	1oz (35 μ m) standard; 2oz for power
Trace impedance 2L	50 Ω microstrip: $W \approx 2.8\text{mm}$ on 1.6mm board (1oz)	Limitation	No continuous GND return – avoid for high-speed
4-LAYER STACKUP (1.6mm total) — JLCPCB JLC04161H-3313			
Layer 1 (Top)	Signal + Components	Thickness Top	0.035mm Cu + 0.1mm prepreg
Layer 2 (In1)	GND plane (solid – reference for Top signals)	Thickness In1	0.035mm Cu + 1.065mm core
Layer 3 (In2)	Power planes (VCC, VCORE, split zones)	Thickness In2	0.035mm Cu + 0.1mm prepreg
Layer 4 (Bot)	Signal + secondary components	Thickness Bot	0.035mm Cu
50Ω microstrip (top)	$W \approx 0.19\text{mm}$ on 0.1mm prepreg to GND (JLC standard)	50Ω stripline	$W \approx 0.14\text{mm}$ embedded between GND layers
6-LAYER STACKUP (1.6mm) — HIGH-SPEED / RF			
L1 Top	Signal (microstrip, ref to L2 GND)	L2	GND (solid plane)
L3	Signal (stripline, ref L2/L4)	L4	Power planes
L5	Signal (stripline, ref L4/L6)	L6 Bottom	Signal + GND pour
IMPEDANCE FORMULAE			
Microstrip Z0	$Z_0 = 87/\sqrt{(\epsilon_r+1.41)} \times \ln(5.98H/(0.8W+T))$	Stripline Z0	$Z_0 = 60/\sqrt{\epsilon_r} \times \ln(4B/(0.67\pi(0.8W+T)))$
W = trace width	H = dielectric height, T = copper thickness	ϵ_r FR4	4.3–4.7 @ 1GHz (use fab's exact Dk)
Differential pair	$Z_{diff} = 2 \times Z_0 \times (1 - 0.347 \times e^{(-2.9 \times S/H)})$	S = gap	Gap between differential pair traces
Common mode Zcc	$Z_{cc} = Z_0/2 \times (1 + 0.347 \times e^{(-2.9 \times S/H)})$	Intra-pair skew	< 5mil mismatch for USB/HDMI/LVDS
JLCPCB 4-LAYER CONTROLLED IMPEDANCE (JLC04161H-3313)			
50Ω microstrip L1	$W = 0.19\text{mm}$ (7.5 mil)	50Ω microstrip L4	$W = 0.19\text{mm}$
50Ω stripline L2	$W = 0.13\text{mm}$ (5.1 mil)	50Ω stripline L3	$W = 0.13\text{mm}$
100Ω diff micro L1	$W = 0.19\text{mm}$, $S = 0.2\text{mm}$	90Ω diff micro	$W = 0.16\text{mm}$, $S = 0.13\text{mm}$ (USB 2.0)
NOTE: Always use your fab's impedance calculator with their specific stackup. JLC, PCBWay, AISLER all publish their layer stackup Dk and thickness values. Request impedance test coupon on panel.			

5. PCB LAYOUT — COMPONENT PLACEMENT STRATEGY

PLACEMENT ORDER & HIERARCHY			
PLACEMENT SEQUENCE (always follow this order)			
1. Connectors	Fix first – constrained by mechanical enclosure / mating	2. Crystals/OSC	Near MCU clock pins, away from switching noise
3. Power ICs	Regulators, converters – near input connector	4. MCU / FPGA	Central, decoupling caps immediately adjacent
5. Signal ICs	ADC, DAC, amplifiers – near their sensor/load	6. Passives	Fill in near their associated active component
7. Bulk capacitors	Near power entry and regulator outputs	8. Test points	Place LAST – fill spare space, accessible locations
CRITICAL PLACEMENT RULES			
Decoupling caps	< 0.5mm from VCC pin, on SAME side as IC if possible	Crystal	< 10mm from MCU pins, symmetric layout, GND guard ring
Diff pair components	Router, USB, HDMI – place so pairs can route parallel	Reset filter	R+C near MCU reset pin – not at connector
Ferrite beads	In series on PWR traces between digital and analogue	ESD protection	At connector entry – first thing signal sees
TVS / protection	< 5mm from connector pin to TVS to GND	Fuse placement	On positive rail, before regulator, near input
THERMAL PLACEMENT			
Hot components	Away from temperature-sensitive parts (oscillators, ADC)	Airflow	Orient components along airflow direction (forced)
Power dissipation	$P_d = (V_{in} - V_{out}) \times I_{out}$ for LDO – use switcher if $P_d > 1W$	Thermal relief	Use spoke thermal connects on THT pads in GND plane
Heatsink clearance	$\geq 3mm$ to adjacent components (or per heatsink spec)	GND via array	Below thermal pad of QFN/DFN – 0.3mm vias, 0.8mm grid
MECHANICAL CONSTRAINTS			
Mounting holes	Keep clearance: 3mm from hole edge to nearest trace	Board edge	Keep 0.2mm min (JLC), prefer 0.5mm from traces/pads
SMD near edge	$\geq 0.5mm$ from board edge for SMD pads	THT near edge	$\geq 1mm$ from board edge for THT pads
Assembly clearance	Leave 5mm around board edge for conveyor grip	Panelisation	V-score: 0.5mm groove each side; break-away 0.3mm holes
NOTE: Placement drives 80% of the board quality. Good placement = easy routing. Bad placement = impossible routing or EMC failures. Spend at least 50% of layout time on placement.			

6. ROUTING TECHNIQUES & TRACE GEOMETRY

TRACE WIDTH vs CURRENT CAPACITY (IPC-2221)			
EXTERNAL TRACES (1oz copper, 10°C rise)			
100 mA	0.10mm (4mil) minimum	500 mA	0.25mm (10mil)
1 A	0.40mm (16mil)	2 A	0.80mm (31mil)
3 A	1.20mm (47mil)	5 A	2.00mm (79mil)
10 A	4.00mm (157mil)	IPC formula	$W = (I / (k \times \Delta T^{0.44}))^{(1/0.725)} / (1.378 \times T^{0.5})$
TRACE WIDTH QUICK RULES			
Signal traces	0.15mm (6mil) min for production, 0.2mm preferred	Power traces	Calculate per IPC-2221, + 50% safety margin

GND fill	Full copper pour — no width limit	High voltage	1mm clearance per 100V (conservative, check standards)
USB D+/D- (90Ω)	W = 0.16mm, Gap = 0.13mm (check fab stackup!)	Ethernet diff	100Ω: W depends on stackup — use fab calculator
ROUTING RULES & GEOMETRY			
45° rule	Use 45° bends, not 90° — reduces EMI/reflections	Curved routing	Smooth arcs — best for RF and high-speed signals
Avoid acute angles	< 45° bends trap acid in etch — design rule violation	Return current	Route signal and its return path close together
No daisy chains	Star topology for sensitive signals (CLK distribution)	T-junctions	Avoid for high-speed — cause reflections
Stub avoidance	Unterminated stubs create resonant reflections	Via placement	Via changes layer — breaks return path continuity
VIA TYPES & SPECIFICATIONS			
Through-via	Drills entire board — cheapest, most common	Blind via	From outer layer to inner layer only (expensive)
Buried via	Inner layers only — highest density, expensive	Micro-via	0.1mm drill, laser — for HDI/BGA escape routing
Via size (JLC std)	Drill: 0.2mm min (0.3mm preferred), pad: 0.4mm	Via-in-pad	Fill + plate for BGA/QFN thermal vias
Via aspect ratio	Board thickness / drill diameter < 10:1 for mechanical	Tented via	Covered with solder mask — prevents solder wicking
RETURN CURRENT & GROUND PLANE			
Return current rule	Return current follows path of LEAST IMPEDANCE (at DC: least resistance; at AC: least inductance = directly under signal trace)		
Plane slot avoidance	Never route signal traces across splits or slots in GND plane — return current must detour → creates EMI loop		
Layer change via	When changing layers: place GND via adjacent to signal via to maintain return path continuity		
Moat around crystal	Flood GND around crystal on all layers, stitch with vias every 1mm, no other traces through crystal keepout		
NOTE: Route from most critical signals first: clocks, diff pairs, high-frequency signals. Leave power and low-speed signals for last. Never compromise signal integrity for routing convenience.			

7. SIGNAL INTEGRITY & HIGH-SPEED DESIGN

SIGNAL INTEGRITY FUNDAMENTALS			
Propagation velocity	$v = c / \sqrt{\epsilon_r} \approx 0.6c$ on FR4 ($\epsilon_r \approx 4.4$)	Delay per inch	≈ 150 ps/inch on FR4 (6 ps/mm)
Critical length	$L_{crit} = T_r / (2 \times T_{pd})$ — where T_r = rise time	Termination	Required when trace length > L_{crit}
Reflection coeff Γ	$\Gamma = (Z_L - Z_0) / (Z_L + Z_0)$	Series term.	$R_s = Z_0 - R_{driver}$ (place near source)
Parallel term.	$R_{term} = Z_0$ (place at load end)	AC termination	$R+C$: $R=Z_0$, $C=100$ pF (blocks DC, terminates AC)
Crosstalk NEXT	Near-end: aggressor energy couples backward	FEXT	Far-end: couples to victim at load end
3W rule	Space traces $3 \times$ trace width (edge-to-edge) to reduce XT	H rule	Keep trace height H above reference plane constant
DIFFERENTIAL PAIR ROUTING			
Match length	Intra-pair length mismatch < 5mil (0.127mm) for USB	Routing style	Route as pair — keep together from source to load
Coupling length	Pairs must be coupled (close) for full differential benefit	Symmetry	Symmetric layout — same vias, same layer changes
USB 2.0	$90\Omega \pm 15\%$, trace length matched ± 100 ps	USB 3.0	90Ω , trace matched ± 5 ps, additional SSC consideration
LVDS pair	100Ω differential, couple $\geq 80\%$ of trace length	HDMI pair	100Ω , 7 differential pairs + HPD + DDC (I2C) + 5V
DECOUPLING STRATEGY (DETAIL)			
Rule 1	100nF on EVERY VCC pin of EVERY IC — no exceptions	Rule 2	Value: 100nF for logic, 1 μ F for analogue/power ICs
Rule 3	Place as close to VCC pin as possible — direct connection	Rule 4	GND side: via directly to GND plane (not long trace)
Rule 5	10 μ F–100 μ F bulk: near regulator output or board entry	Rule 6	Use X7R/X5R dielectric — NOT Y5V/Z5U (capacitance drops)
PDN impedance	Target < 10m Ω across frequency band of interest	Resonance	Place caps with different values to fill frequency gaps
CLOCK SIGNAL ROUTING			
Series resistor	22–33 Ω series R at source — reduces ringing and EMI	Load cap	Each receiver ≈ 5 –15pF — keep total load low
Point-to-point	Single source to single load — no branches	Daisy chain clk	Only if using differential LVDS/LVPECL
Clock buffer	Use for >2 loads — fan-out buffer (e.g. SY100EL16)	Clock distribution	Star topology from buffer to all receivers
Guard traces	GND traces adjacent to clock trace on same layer	GND vias	Stitch vias alongside clock trace every 5mm
NOTE: Signal integrity is physics — it cannot be fixed by firmware. Always simulate critical nets with HyperLynx/QUCS/Sigrity or use rule-based checking (rise time vs trace length) before layout is finalized.			

8. POWER DISTRIBUTION NETWORK (PDN) & PLANES

POWER PLANE DESIGN			
PLANE vs TRACE POWER DISTRIBUTION			
Use plane when	Supply current > 3A, or many loads distributed across board	Use trace when	< 1A, localized load, thin board or 2-layer

Split planes	Split inner plane into power domains (VCC/VCORE/VDDA)	Split rule	Never route signal traces across a plane split
Plane capacitance	$C_{pln} = \epsilon_0 \times \epsilon_r \times A / d$ – inner layer pair is a capacitor	Benefit	PDN capacitance: ~100pF/cm² for 0.1mm dielectric
Plane inductance	Very low inductance – much better than wide traces	GND plane	Dedicate at least ONE full layer to solid GND
VOLTAGE REGULATOR PLACEMENT & LAYOUT			
LDO layout	Vin cap → LDO → Vout cap → load (linear path)	LDO feedback R	Feedback resistors: short traces, close to VOUT pin
Buck converter	Input cap → Switch → Inductor → Output cap (hot loop small!)	Hot loop	Vin cap, switch FETs, inductor in tight triangle
Boost converter	GND→Inductor→Switch→Diode→Output cap → load	CCM vs DCM	CCM: lower ripple; DCM: enters at light load
Switcher GND	Separate switcher GND to star point – away from analogue	EMI reduction	Snubber RC across diode; shield inductor; spread spectrum
CURRENT PATH & HOT LOOP MINIMISATION			
Hot loop	The loop formed by high dI/dt switching – minimize its area: Input Cap → MOSFET High → MOSFET Low → Input Cap GND. Keep all three as tight triangle.		
$L = \mu_0 \times A / h$	Loop inductance proportional to area A – halve loop area → halve inductance → halve switching voltage spike		
Via inductance	Each via ≈ 0.5-1nH – use multiple vias in parallel for high-current paths (I > 1A per via is excessive)		
PLANE STITCHING VIAS			
GND stitch spacing	Every $\lambda/20$ at highest frequency (< 5mm for 1GHz)	Around signals	Row of GND vias alongside differential pairs
Layer transition	GND via next to every signal via – maintain return path	Board edge	Ring of GND vias 1mm inside board perimeter
Power stitching	Multiple vias for power layer connections (≥ 4 vias/A)	Thermal vias	0.3mm drill, 0.2mm wall, array under thermal pads
NOTE: PDN design: target < 10mΩ impedance at the IC's power pins from DC to 1GHz. Use a PDN analyzer tool (Altium PDN Analyzer, SIwave) for complex designs. Rule of thumb: 1 decap per 100MHz of bandwidth.			

9. EMC / EMI PCB DESIGN RULES

EMC DESIGN FUNDAMENTALS			
EMI source	Any fast-switching signal (clock, PWM, switcher)	EMI coupling	Radiated (electric/magnetic loop) or conducted
Electric field (E)	Driven by high dV/dt – reduce by termination, shielding	Magnetic (H)	Loop area × current – minimize switching loop area
Rise time vs freq	Highest harmonic ≈ 0.35 / Tr (Tr = 10-90% rise time)	3ns rise time	≈ 117MHz content → needs SI/EMC treatment above 100MHz
EMC LAYOUT TECHNIQUES			
GND plane	Solid GND plane is the single biggest EMC improvement	No slot in GND	Every slot or gap disrupts return currents → radiates
Partition board	Analogue section Digital section Power section – separate	Single star GND	Connect partitions at single low-impedance point
Filter at entry	Every cable/connector: ferrite + TVS + CM choke at entry	Pi-filter	C → L → C on power entry – handles both CM and DM
Spread spectrum	Enable in MCU/switcher – spreads EMI over wider frequency	Shielding can	Metal shield over RF section, gasketed if needed
Guard ring	Driven GND ring around sensitive analogue section	Keepout zone	Exclude switchers/clock from analogue section
COMMON MODE & DIFFERENTIAL MODE FILTERING			
Common mode noise	Both conductors swing together relative to GND	DM noise	Conductors swing opposite – normal signal mode
CM choke	High impedance to CM, transparent to DM	CM cap (Cy)	Line-to-GND: 1-10nF (safety rated Y1/Y2 for mains)
DM inductor	Series on each line – with Cx cap for pi-filter	Cx cap	Line-to-line capacitor: safety rated X1/X2 for mains
Ferrite bead	Resistive at high frequency – not a pure inductor	Ferrite select	ZF at 100MHz > 600Ω for digital; plot impedance vs freq
PCB SHIELDING & GROUNDING FOR EMC			
Chassis GND	Connect PCB GND to chassis at ONE point (star GND)	Strap bond	Low-impedance strap < λ/20 at highest frequency
Slot antenna	Exposed signals on board edge radiate like slot antennas	Mitigation	Keep signals 3mm from board edge; add GND pour
Cable shield	Terminate cable shield at board entry via low-Z cap/direct	Pigtail	< 25mm pigtail – longer pigtail = antenna above 100MHz
EMC STANDARDS (SA / International)			
CISPR 32	Multimedia equipment emissions (replaces EN 55032)	CISPR 35	Multimedia equipment immunity
IEC 61000-4-2	ESD immunity: ±2kV contact, ±4kV air	IEC 61000-4-4	EFT (fast transient): ±2kV on power lines
IEC 61000-4-5	Surge: ±2kV L-L, ±4kV L-GND	ICASA / SABS	SA: ICASA type approval required for wireless products
NOTE: Get EMC right in layout — post-layout EMC fixes (ferrites, caps) add cost and complexity. A solid GND plane + proper partitioning + decoupling solves 80% of EMC issues before testing.			

10. DRC / DFM / MANUFACTURING RULES

DESIGN RULE CHECK (DRC) PARAMETERS
JLCPCB STANDARD PROCESS MINIMUMS

Min trace width	0.09mm (3.5mil) — prefer 0.127mm (5mil)	Min clearance	0.09mm (3.5mil) trace-to-trace; prefer 0.127mm
Min drill (mech)	0.2mm drill, 0.4mm pad — prefer 0.3mm drill	Min drill (laser)	0.1mm micro-via
Min annular ring	0.13mm (5mil) — outer layer	Min hole/pad	Pad = drill + 2×annular ring + 2×fab tolerance
Soldermask web	0.1mm between adjacent SMD pads (preferred 0.15mm)	Silkscreen width	0.1mm min line width; 0.8mm min text height
Board edge clearance	0.2mm min (trace/copper to board edge)	V-score	Material remaining: 0.4mm (JLC standard V-score)

DESIGN FOR MANUFACTURE (DFM) CHECKLIST

Fiducial marks	3× minimum on PCB for pick & place alignment	Fiducial size	1mm Cu circle, 3mm solder-mask clearance, on top copper
Tooling holes	2× 2.0mm or 3.0mm (NPTH) for PCB holding during reflow	Paste coverage	Stencil thickness 0.12mm for 0402, 0.15mm for 0603+
Assembly polarity	Polarity marker on every polarised part in silkscreen	Pin 1 marker	Triangle or dot on IC silkscreen — must be visible
Part clearance	≥ 0.3mm between adjacent SMD parts (reflow)	THT clearance	≥ 1mm between THT parts; solder side clearance for wave
Solder bridge risk	Fine-pitch (<0.65mm) needs solder mask between pads	Tomb-stoning	Match pad size/length exactly for small passives

DESIGN FOR ASSEMBLY (DFA) BEST PRACTICES

One-sided SMD	Reduce cost — all SMD on top, THT on bottom if possible	Consistent orient	Orient all passives same direction — speeds visual check
Avoid bottom SMD	Saves second reflow pass — significant cost saving	Large parts last	Tall parts (caps, connectors) placed after small SMD
Thermal relief	THT pads in GND pour — use 4-spoke thermal relief	SMD in pour	Remove thermal relief for SMD pads in pour (better solder)

GERBER OUTPUT CHECKLIST

Gerber layers	F.Cu (GTL) B.Cu (GBL) In1.Cu (G1) In2.Cu (G2) F.Paste (GTP) B.Paste (GBP) F.Silkscreen (GTO) B.Silkscreen (GBO) F.Mask (GTS) B.Mask (GBS) Edge.Cuts (GKO)
Drill files	PTH drill (DRL) + NPTH drill (DRL) + Drill map (PDF or DXF) — use Excellon format
Assembly files	Centroid/CPL (CSV: reference, x, y, rotation, side) + BOM (CSV: ref, qty, value, MPN, footprint)

NOTE: Always visually inspect Gerber files in a viewer (KiCAD Gerber viewer, Gerber Panelizer, GerbView) before ordering. Check drill alignment, silkscreen completeness, and mask openings.

11. KICAD — INTERFACE & GLOBAL WORKFLOW

KICAD 7/8 PROJECT STRUCTURE			
*.kicad_pro	Project file – links all files together	*.kicad_sch	Schematic (main sheet or top-level hierarchy)
*.kicad_pcb	PCB layout file	*.kicad_sym	Symbol library file
*.kicad_mod	Footprint library file (folder = .pretty)	fp-info-cache	Footprint info cache – regenerate if corrupt
sym-lib-table	Symbol library table – maps library names to paths	fp-lib-table	Footprint library table – same for footprints
Backups/	Auto-backup folder – KiCAD saves every N minutes	*.kicad_wks	Page layout (title block) worksheet file
PROJECT WORKFLOW IN KICAD			
1. New project	File → New Project → creates .kicad_pro + blank .kicad_sch + blank .kicad_pcb		
2. Draw schematic	Open Eeschema (schematic editor) → add symbols → draw wires → assign footprints → run ERC		
3. Sync to PCB	In Eeschema: Tools → Update PCB from Schematic (F8) → Review changes → Update		
4. Layout	Open Pcbnew → set board outline → place components → set design rules → route → fill zones → DRC		
5. Output	File → Fabrication Outputs → Gerbers, Drill Files, BOM, PnP (centroid)		
KICAD SETTINGS TO CONFIGURE FIRST			
Preferences → Libraries	Set global symbol & footprint library paths	Scripting console	View → Scripting Console – Python 3 REPL for automation
Origin point	Place → Grid Origin at corner of board – sets (0,0)	Grid settings	Fine: 0.1mm or 0.05mm; Coarse: 1mm; use G key to toggle
Design rules	File → Board Setup → Design Rules – set ALL clearances	Net classes	Board Setup → Net Classes – assign width/clearance per net
Colour scheme	Preferences → Manage Colour Themes – import community theme	Interactive router	Settings → Interactive Router: Highlight Mode, Push & Shove
NOTE: Always use a project-local library (MyProject.kicad_sym / MyProject.pretty) for custom symbols and footprints. Never edit global libraries — updates will overwrite your changes.			

12. KICAD SCHEMATIC EDITOR — SHORTCUTS & TRICKS

Shortcut	Action	Context
NAVIGATION		
Scroll wheel	Zoom in/out	<i>Schematic</i>
Middle click drag	Pan view	<i>Schematic</i>
Ctrl+Home	Fit entire schematic in view	<i>Schematic</i>
F	Zoom to fit selection	<i>Schematic</i>
ADDING & EDITING COMPONENTS		
A	Add symbol (opens symbol chooser)	<i>Schematic</i>
P	Add power symbol (VCC, GND, etc.)	<i>Schematic</i>
W	Draw wire	<i>Schematic</i>
Q	End wire / end drawing action	<i>Schematic</i>
L	Add net label	<i>Schematic</i>
E	Edit properties of selected item	<i>Schematic</i>
R	Rotate selected item (90° CCW)	<i>Schematic</i>

X	Mirror horizontally (flip X)	<i>Schematic</i>
Y	Mirror vertically (flip Y)	<i>Schematic</i>
G	Grab item (drag while maintaining connections)	<i>Schematic</i>
M	Move item (disconnect from wires)	<i>Schematic</i>
Del	Delete selected item	<i>Schematic</i>
Ctrl+D	Duplicate selected item	<i>Schematic</i>
Ctrl+Z / Ctrl+Y	Undo / Redo	<i>Schematic</i>
ANNOTATION & TOOLS		
T	Add text / label annotation	<i>Schematic</i>
K	No Connect flag (X marker on pin)	<i>Schematic</i>
J	Add junction (dot where wires cross and connect)	<i>Schematic</i>
Ctrl+A	Annotate schematic (auto-number all components)	<i>Tools menu</i>
Ctrl+Shift+A	Clear annotation (reset all reference designators)	<i>Tools menu</i>
Ctrl+L	Edit symbol fields (mass edit all component properties)	<i>Tools menu</i>
Ctrl+F	Find / search for component or net	<i>Schematic</i>
F8	Update PCB from Schematic (sync to layout)	<i>Tools menu</i>
Ctrl+R	Run ERC (Electrical Rules Check)	<i>Tools menu</i>
HIERARCHY & SHEETS		
S	Add hierarchical sheet (sub-schematic block)	<i>Schematic</i>
I	Add hierarchical label (port on sub-sheet)	<i>Schematic</i>
Ctrl+click	Navigate into/out of hierarchical sheet	<i>Schematic</i>
Ctrl+Shift+U	Highlight net (highlight all connected wires & pins)	<i>Schematic</i>

KiCAD Schematic — Pro Tips & Tricks

```
// TIP 1: Net Inspector — Inspect → Net Inspector
//   Shows every net, its connections, and flags disconnected nets

// TIP 2: Symbol from CSV — Use KiCAD's symbol generator scripts
//   Tools → Edit Symbol → Import from Datasheet PIN table (KiCAD 8)

// TIP 3: Bus notation for multi-bit signals
//   Draw bus line, label: DATA[0..7] — connects DATA0..DATA7 auto

// TIP 4: Global vs Local labels
//   Net Label (L) = local to current sheet
//   Global Label = visible across ALL sheets in project
//   Power Symbol (P) = special global — always use for VCC/GND

// TIP 5: Assign footprints BEFORE layout — use footprint browser
//   Ctrl+click on component → Properties → Footprint → Browse
//   Or use Assign Footprints tool (Tools → Assign Footprints)

// TIP 6: ERC customisation — Tools → ERC → ERC Violations
//   Right-click violation → Ignore rule to suppress false positives
//   Document WHY you ignored — add schematic note nearby

// TIP 7: Highlight net across PCB and Schematic simultaneously
//   Click net name/label → Inspect → Highlight Net
//   Works cross-application (both editors open simultaneously)
```

13. KICAD PCB EDITOR (Pcbnew) — SHORTCUTS & TRICKS

Shortcut	Action	Context
NAVIGATION		
Scroll wheel	Zoom in/out	PCB
Middle drag	Pan view	PCB
Ctrl+Home	Zoom to fit board	PCB
` (Backtick key)	Toggle ratsnest visibility	PCB
Ctrl+backtick	Toggle track and via visibility	PCB
ROUTING		
X	Start routing trace from pad/via (interactive router)	PCB
V	Add via while routing (changes layer + adds via)	PCB
/	Switch to next allowed copper layer while routing	PCB
W	Change trace width while routing (cycles Net Classes)	PCB
U	Expand selection to entire connected track segment	PCB
I	Select all tracks in net	PCB
Q	End routing / exit routing mode	PCB
Backspace	Undo last routing step (shorten trace)	PCB while routing
D	Interactive router: drag track (extends/reshapes)	PCB
PLACEMENT & EDITING		
E	Edit properties of selected item	PCB
R	Rotate selected item (use in properties for exact angle)	PCB
F	Flip selected item to other side of board	PCB
G	Drag item while maintaining connections	PCB
M	Move item	PCB
T	Add text	PCB
Ctrl+D	Duplicate selected item	PCB
Ctrl+G	Group selected items	PCB
Ctrl+Shift+G	Ungroup items	PCB
ZONES (COPPER POUR)		
Z	Draw filled zone (copper pour area)	PCB
Ctrl+B	Fill ALL zones (copper pour fill/update)	PCB
Ctrl+Shift+B	Un-fill / remove copper pour fill from all zones	PCB
E on zone	Edit zone properties (net, clearance, fill, priority)	PCB
INSPECTION & TOOLS		
Ctrl+Shift+U	Highlight net across PCB and schematic	PCB
Ctrl+R	Run DRC (Design Rules Check)	PCB
Inspect → Net Inspector	See all nets, lengths, connected components	Menu
Inspect → Board Statistics	Track count, via count, pad count summary	Menu
Alt+3	Open 3D viewer (with live sync to PCB changes)	PCB
Ctrl+Shift+M	Measure distance between two points	PCB
LAYER MANAGEMENT		
Layer selector	Top toolbar dropdown — select active layer	PCB
+ / -	Cycle to next/previous copper layer	PCB
Ctrl+Click layer	Set active layer in layer panel	PCB
; (semicolon)	Switch to component layer (auto: top or bottom based on footprint)	PCB

14. KICAD — EFFICIENCY TRICKS & ADVANCED FEATURES

KICAD POWER USER TECHNIQUES			
INTERACTIVE ROUTER MODES			
Push & Shove	Router pushes existing traces out of way (best mode)	Walk Around	Router avoids existing traces – does not move them
Highlight Collision	Shows DRC errors in red as you route – visual feedback	Enable mode	Route → Interactive Router Settings → Route Mode
Lock track	Lock track → router cannot move it in Push & Shove	Track locking	Select track → right-click → Lock
Diff pair router	Route → Route Differential Pair (D+D-) as matched pair	Dp skew tuning	Route → Tune Differential Pair Skew (length matching)
Length tuning	Route → Tune Track Length → add meanders to match length	Accordion	Meander style: adds serpentine section for length matching
DESIGN RULE MANAGER			
Net class rules	Board Setup → Net Classes → Add Class → set Width/Clearance/Via Size per net class (assign nets in class)		
Custom DRC rule	(rule high_speed_clearance (condition "A.NetClass == 'HS'") (constraint clearance (min 0.3mm)))		
Diff pair rule	(rule diff_pair_gap (condition "A.isDiffPair()") (constraint diff_pair_gap (min 0.13mm) (max 0.13mm)))		
Via fence rule	(rule no_vias_in_keepout (condition "A.Type == 'Via'") (constraint disallow via))		
FOOTPRINT EDITOR TRICKS			
Pad numbering	Select multiple pads → right-click → Renumber Pads	Pad properties	Set paste/mask expansion per pad (override global)
3D model	Footprint Properties → 3D Models → Add .step/.wrl file	3D offset	Adjust X/Y/Z offset and rotation to align model
Courtyard check	Tools → Check Footprints → reports missing courtyard	Copy from lib	Footprint Editor → Load Footprint from Library → edit
Pattern from array	Edit → Create Array (Ctrl+T) – creates pad/via grid	Mirror pads	Select pads → right-click → Mirror X / Mirror Y
SCRIPTING & AUTOMATION			
Open console	View → Scripting Console → Python 3 REPL (or pcbnew → open from command line)		
Enumerate nets	<pre>import pcbnew; b=pcbnew.GetBoard(); [print(n.GetNetname()) for n in b.GetNets().values()]</pre>		
Move component	<pre>ref = b.FindFootprintByReference('U1'); ref.SetPosition(pcbnew.FromMM(50, 30))</pre>		
Run DRC in script	<pre>import pcbnew; drc=pcbnew.WriteDRCReport(b, '/tmp/drc.txt', pcbnew.EDA_UNITS_MILLIMETRES, True)</pre>		
Set trace width	<pre>for t in b.GetTracks(): t.SetWidth(pcbnew.FromMM(0.2)) if t.GetNetname()=='GND' else None</pre>		
PLUGINS — MUST-HAVE KICAD PLUGINS			
KiBuzzard	Draw complex labels/logos in PCB silkscreen	Footprint Wizard	Built-in: SMD pads, connectors, QFN patterns
Interactive HTML BOM	Clickable BOM linked to board view – easier assembly	PCBWay plugin	Direct Gerber export + ordering from PCBWay
KiKit	Panel generation (array boards with mouse bites/V-score)	Freerouter	External Java auto-router (import/export DSN)
PCBDraw	Renders PCB to SVG/PNG for documentation	fab_outputs	Automates Gerber + drill + BOM export in one click

NOTE: Install plugins via: PCB Editor → Plugin and Content Manager (PCM). KiKit and Interactive HTML BOM are essential for professional workflows. Bookmark [kicad.org/download/plugins/](https://www.kicad.org/download/plugins/) for the full catalogue.

15. KICAD — SYMBOL & FOOTPRINT LIBRARIES

LIBRARY MANAGEMENT BEST PRACTICES			
LIBRARY TYPES			
Global library	Available to ALL projects – set in Preferences → Manage Libraries	Project library	Stored in project folder – portable, recommended
KiCAD std libs	kicad-symbols (GitHub) – regularly updated	Community libs	SnapEDA, UltraLibrarian, Component Search Engine
samacsys plugin	Imports symbols+footprints from SamacSys directly in KiCAD	LCSC	EasyEDA parts → convert via easyeda2kicad tool
CREATING A PROJECT LIBRARY			
Symbol lib	Symbol Editor → File → New Library → Save as MyProject.kicad_sym → add to project lib table		
Footprint lib	Footprint Editor → File → New Library → Save as MyProject.pretty → add to fp-lib-table		
sym-lib-table	(lib (name "MyProject") (type "KiCad") (uri "\${KIPRJMOD}/MyProject.kicad_sym") (options "") (descr ""))		
fp-lib-table	(lib (name "MyProject") (type "KiCad") (uri "\${KIPRJMOD}/MyProject.pretty") (options "") (descr ""))		
SYMBOL CREATION CHECKLIST			
Pin names	Match EXACTLY to datasheet – case sensitive	Pin numbers	Match datasheet pin numbering – critical for footprint mapping
Pin types	Input/Output/Bidirectional/Power In/Power Out/Passive	Hidden pins	VCC/GND supply pins: hide + assign to power net
Reference prefix	U for IC, R for resistor, C for cap, etc.	Pin length	100 mil (2.54mm) standard pin length
Body units	Multi-unit for dual/quad op-amps – share power pins	Datasheet link	Add datasheet URL in component properties field
FOOTPRINT CREATION CHECKLIST			
IPC calculator	Use KiCAD built-in IPC-7351 footprint wizard	Courtyard	0.25mm outside body (production) / 0.5mm (hand solder)
Fab layer	Exact body outline from datasheet drawing	Silkscreen	Simplified body + pin 1 dot – must not overlap pads
3D model	Download STEP from manufacturer, align precisely	Check	Footprint Editor → Tools → Check Footprints
EASYEDA → KICAD IMPORT			
easyeda2kicad	pip install easyeda2kicad → easyeda2kicad --lcsc-id C14663 --output MyLib (imports by LCSC part number)		
LCSC to KiCAD	Search LCSC part → click EasyEDA symbol → in KiCAD Plugin: EasyEDA Pro Import → converts symbol+footprint+3D		
NOTE: ALWAYS verify imported symbols and footprints against actual datasheets. Automated imports often have wrong pin types, missing courtyard, or incorrect 3D orientation. Never use unverified parts in production designs.			

16. KICAD — PYTHON SCRIPTING RECIPES

Useful Python Scripts for KiCAD Automation

```
import pcbnew, os

# — Get board object —————
b = pcbnew.GetBoard()           # from within KiCAD console
b = pcbnew.LoadBoard('/path/to/file.kicad_pcb') # from external script
```

```

# — List all footprints —————
for fp in b.GetFootprints():
    print(fp.GetReference(), fp.GetValue(), fp.GetPosition())

# — Find a specific footprint —————
u1 = b.FindFootprintByReference('U1')
print(u1.GetPosition())           # returns (x, y) in IU

# — Move component to specific XY (mm) —————
u1.SetPosition(pcbnew.VECTOR2I(pcbnew.FromMM(50.0), pcbnew.FromMM(30.0)))

# — Set component orientation (degrees) —————
u1.SetOrientationDegrees(90)      # 0, 90, 180, 270

# — Set trace width for all GND traces —————
for track in b.GetTracks():
    if track.GetNetname() == 'GND':
        track.SetWidth(pcbnew.FromMM(0.5))

# — Add via at position —————
via = pcbnew.PCB_VIA(b)
via.SetPosition(pcbnew.VECTOR2I(pcbnew.FromMM(25), pcbnew.FromMM(25)))
via.SetDrill(pcbnew.FromMM(0.3))
via.SetWidth(pcbnew.FromMM(0.6))
via.SetNet(b.FindNet('GND'))
b.Add(via)

# — Generate Gerbers from script —————
plot_ctrl = pcbnew.PLOT_CONTROLLER(b)
plot_opts = plot_ctrl.GetPlotOptions()
plot_opts.SetOutputDirectory('/output/gerbers/')
plot_opts.SetPlotFrameRef(False)
plot_opts.SetPlotValue(True)
plot_opts.SetPlotReference(True)
layers = [(pcbnew.F_Cu, 'F_Cu'), (pcbnew.B_Cu, 'B_Cu'),
          (pcbnew.F_Silks, 'F_Silkscreen'), (pcbnew.Edge_Cuts, 'Edge_Cuts')]
for layer, name in layers:
    plot_ctrl.SetLayer(layer)
    plot_ctrl.OpenPlotfile(name, pcbnew.PLOT_FORMAT_GERBER, name)
    plot_ctrl.PlotLayer()
plot_ctrl.ClosePlot()

# — Export drill file —————
drill_writer = pcbnew.EXCELLON_WRITER(b)
drill_writer.SetOptions(False, True, b.GetDesignSettings().GetAuxOrigin(), True)
drill_writer.CreateDrillandMapFilesSet('/output/gerbers/', True, False)

# — Save board —————
b.Save(b.GetFileName())
pcbnew.Refresh()                 # refresh PCB editor view

```

KiCAD Custom DRC Rules File (.kicad_dru)

```

# Place in project root as: project_name.kicad_dru
# Board Setup → Design Rules → Load Rules File

(version 1)

# — High-speed net class clearance —————
(rule hs_clearance
  (condition "A.NetClass == 'HighSpeed' && B.NetClass == 'HighSpeed'")
  (constraint clearance (min 0.3mm)))

# — Differential pair gap constraint —————
(rule diff_pair
  (condition "A.isDiffPair()")
  (constraint diff_pair_gap (min 0.127mm) (max 0.127mm))
  (constraint diff_pair_uncoupled (max 5mm)))

```

```
# — Crystal keepout — no vias within 3mm —————
(rule crystal_keepout
  (condition "A.Type == 'Via' && B.Reference == 'Y1'")
  (constraint clearance (min 3mm)))

# — Power trace minimum width —————
(rule power_trace_width
  (condition "A.NetClass == 'Power'")
  (constraint track_width (min 0.5mm)))

# — No traces in courtyard of U1 —————
(rule u1_courtyard_keepout
  (condition "A.Type == 'Track' && B.Reference == 'U1'")
  (constraint courtyard_clearance (min 0.5mm)))

# — Minimum annular ring —————
(rule annular_ring
  (constraint annular_width (min 0.15mm)))
```

17. HIGH-SPEED & RF PCB DESIGN RULES

HIGH-SPEED INTERFACE ROUTING RULES			
USB 2.0 (480Mbps)			
Impedance	90Ω ±15% differential	Trace width	W = 0.16mm, Gap S = 0.13mm (verify with fab stackup)
Max length	18 inches (457mm) on PCB from connector to IC	Series resistor	22–33Ω on each line at source – dampen reflections
Routing	Route as tightly coupled pair, no vias if possible	ESD protection	PRTR5V0U2X or similar near connector before series R
Crystal	12MHz or 48MHz – within 10mm of USB controller	Ferrite	600Ω@100MHz ferrite on VBUS before 100nF bypass cap
USB 3.0 / USB 3.1 (5/10 Gbps)			
Impedance	85Ω ±10% differential SuperSpeed pairs	Length matching	SuperSpeed pair match: ±1mm intra-pair, ±5mm inter-pair
No vias	Vias add capacitive discontinuity – avoid on SS pairs	Layer changes	If via needed: back-drill to remove stub; compensate spacing
SMA connector	50Ω SMA to coax → 100Ω diff pair transition carefully	Crosstalk	≥ 20mil spacing between TX and RX pair groups
DDR SDRAM (DDR3/DDR4)			
Topology	Fly-by topology (DDR3/4) – not star (DDR2 and earlier)	Termination	On-die termination (ODT) in DDR3/4 – no external R
Length match (DQ)	±25mil within byte lane (DQ to DQS)	Address match	±50mil for address/command lines
Via rule DDR4	Avoid vias on data lines – back-drill if unavoidable	Impedance DDR3	50Ω single-ended, 100Ω differential (DQS pairs)
Decoupling DDR	100nF per VDD/VDDQ pin + 10μF bulk per power domain	Guard vias	Row of GND vias between DDR bus and other circuits
RF PCB RULES (Sub-GHz to 6GHz)			
Microstrip 50Ω	Calc W from ϵ_r , H, T using fab stackup data	Ground plane	Solid GND on layer directly below RF trace – no splits
RF keepout	No traces, vias, or fills within $\lambda/10$ of antenna	Antenna isolation	20dB min isolation between TX and RX antenna ports
Component placement	RF IC → matching network → antenna – linear path	Balun	Place between unbalanced IC and balanced antenna
SMA edge launch	Tab on SMA: must connect to GND plane immediately	Via fence	Row of GND vias alongside 50Ω trace every $1/10 \lambda$
Component lead length	Keep leads $< \lambda/20$ for RF passives (use 0402 or 0201)	NF connector	U.FL/IPEX for internal RF – straight vertical mount
ETHERNET (10/100/1000BASE-T)			
Isolation transformer	Magnetics mandatory – check if built into RJ-45 or separate	Common mode	1500V isolation in magnetics per IEEE 802.3
PoE protection	TVS + CM choke on data pairs before transformer	Impedance	100Ω differential for 1000BASE-T pairs
Centre tap	Magnetics centre tap connects to 75Ω to chassis GND	Bob Smith term	75Ω to GND on each pair at ICM – reduces EMI
NOTE: For RF designs: always run 3D EM simulation (EMPro, HFSS, Sonnet, or free Scikit-RF+OpenEMS) before taping out. Rules of thumb fail at GHz frequencies — simulation is mandatory.			

18. THERMAL MANAGEMENT IN PCB DESIGN

THERMAL CALCULATIONS

Junction temp T_j	$T_j = T_a + P_d \times (\theta_{ja}) \quad [^{\circ}\text{C}]$	With heatsink	$T_j = T_a + P_d \times (\theta_{jc} + \theta_{cs} + \theta_{sa})$
θ_{ja} (junction-air)	IC datasheet – no heatsink, free air	θ_{jc} (jctn-case)	IC datasheet – best case (with heatsink)
θ_{cs} (case-sink)	Thermal interface material: 0.5–2 $^{\circ}\text{C}/\text{W}$	θ_{sa} (sink-air)	Heatsink spec – function of airflow
PCB copper θ	Cu layer: $\theta \approx 0.003^{\circ}\text{C}/\text{W}$ per mm^2 per layer	FR4 θ	FR4: 0.3 $\text{W}/\text{m}\cdot\text{K}$ (poor) – copper spreads heat
TIM (thermal paste)	3–8 $\text{W}/\text{m}\cdot\text{K}$ typical; phase-change 3–5 $\text{W}/\text{m}\cdot\text{K}$	Thermal pad	Graphite 5–10 $\text{W}/\text{m}\cdot\text{K}$; indium foil 80 $\text{W}/\text{m}\cdot\text{K}$

PCB THERMAL DESIGN TECHNIQUES

Thermal vias	Array of 0.3mm vias under QFN/DFN thermal pad	Via spacing	0.8–1.0mm grid – fills with solder during reflow
Copper pour	Large GND pour on bottom layer – spreads heat laterally	Heavy copper	2oz (70 μm) or 3oz for power stages – doubles Cu thickness
Component spacing	Hot components away from oscillators, sensors	Airflow orient	Long axis of hot components parallel to airflow
Edge connectors	Connector bodies block airflow – leave gap beside them	Fan placement	Exhaust away from sensitive analogue section

QFN / DFN THERMAL PAD DESIGN

Exposed pad area	Match PCB pad to IC package thermal pad exactly (from datasheet) – do NOT extend beyond package outline
Paste coverage	Cover 50–70% of thermal pad area with stencil apertures (cross-hatch or 9-circle pattern)
Via fill	Via-in-pad: fill with copper fill, plate over – prevents solder wicking into vias (required for small pads)
Thermal resistance	$R_{\text{thermal_vias}} = R_{\text{thermal_FR4}} / N_{\text{vias}}$ – each via is parallel thermal path through board

NOTE: Use thermal simulation (Ansys Icepak, SolidWorks Flow, or free tools like OpenFOAM) for high-power designs. For quick checks: online calculators at Analog Devices or TI thermal design tools.

19. KICAD — COMPLETE DESIGN RULES SETUP

BOARD SETUP → DESIGN RULES (File → Board Setup)			
CONSTRAINTS (minimum values — for JLCPCB standard process)			
Min clearance	0.127mm (5mil) — preferred, use 0.09mm absolute minimum	Min track width	0.127mm (5mil) — use 0.09mm for escape routing only
Min via hole	0.3mm drill (0.6mm pad) for standard; 0.2mm absolute	Min via annular	0.13mm (5mil)
Min uVia hole	0.1mm (laser) — only for HDI stackup	Min uVia pad	0.2mm
Min hole spacing	0.254mm drill-to-drill spacing	Min silkscreen	0.1mm line width; 0.8mm text height
Copper to edge	0.2mm absolute; prefer 0.5mm	Drill to edge	0.3mm absolute; prefer 0.5mm
NET CLASSES SETUP			
Default class	Width: 0.2mm Clearance: 0.2mm Via drill: 0.3mm Via pad: 0.6mm μ Via drill: 0.1mm μ Via pad: 0.2mm		
Power class	Width: 0.5mm Clearance: 0.3mm Via drill: 0.4mm Via pad: 0.8mm — assign VCC, GND, 3V3, 5V nets		
HighSpeed class	Width: 0.15mm Clearance: 0.25mm Via drill: 0.2mm Via pad: 0.4mm — assign USB, CLK, SPI nets		
DiffPair class	Width: 0.15mm Clearance: 0.25mm Gap: 0.127mm — assign USB_D+/D-, ETH pairs		
LAYER SETUP			
Copper layers	2, 4, 6, 8... — set in Board Setup → Board	Board thickness	1.6mm standard; 0.8mm, 1.0mm, 2.0mm available at JLC
Copper weight	1oz (35 μ m) default; 2oz (70 μ m) for power layers	Dielectric	FR4 standard; Rogers/Taconic for RF > 3GHz
Layer naming	F.Cu=Top, B.Cu=Bottom, In1.Cu..InN.Cu for inner layers	Layer purpose	Define in layer name: In1.Cu_GND, In2.Cu_Power
SOLDER MASK & PASTE SETTINGS			
Solder mask clearance	0.05mm global — increased locally for fine-pitch via tenting	Paste margin	0.00mm global (match pad exactly) — adjust in footprint
Mask min web width	0.1mm between adjacent pads (JLC minimum)	Via tenting	Tented vias: mask over via (no opening) — check with fab
HASL vs ENIG	HASL: cheap, uneven; ENIG: flat, expensive, needed for QFN	OSP	Organic Solderability Preservative — flat, low cost
NOTE: Import your fab's design rule file if available (JLC provides .kicad_dru files). Validate with their online DFM checker after Gerber export. Set up net classes BEFORE routing — changing width mid-route is painful.			

20. KICAD — GERBER EXPORT & FAB ORDERING

GERBER EXPORT SETTINGS (KiCAD 7/8)			
PLOT SETTINGS (File → Fabrication Outputs → Gerbers)			
Format	Gerber X2 (preferred) or RS-274X	Coordinate format	4:6 (mm) — maximum precision
Plot footprint values	ON — for assembly reference	Plot ref desig.	ON — for assembly reference
Force plotter units	mm (metric)	Check zones	Enable 'Check zone fills before plotting'
Layers to plot	All copper + mask + paste + silkscreen + edge cuts + courtyard	Separate files	One Gerber file per layer (standard practice)
LAYERS → FILE MAPPING (JLC naming preference)			

F.Cu → GTL	Front copper (top layer)	B.Cu → GBL	Back copper (bottom layer)
In1.Cu → G1L	Inner copper layer 1	In2.Cu → G2L	Inner copper layer 2
F.Mask → GTS	Front solder mask	B.Mask → GBS	Back solder mask
F.Paste → GTP	Front paste (stencil)	B.Paste → GBP	Back paste
F.Silkscreen → GTO	Front silkscreen	B.Silkscreen → GBO	Back silkscreen
Edge.Cuts → GKO	Board outline (mechanical)	Drill → DRL	Excellon drill file (PTH + NPTH separate)
DRILL FILE EXPORT (File → Fabrication Outputs → Drill Files)			
Format	Excellon – universally accepted	Units	Millimetres; 4:4 coordinate format
Zeros format	Suppress leading zeros	Separate files	PTH (plated) and NPTH (non-plated) separate files
Map file	Export drill map as PDF or DXF for visual check	Oval holes	Use Alternate Drilling mode for oval slots
ASSEMBLY (PNP) EXPORT			
Centroid file	File → Fabrication Outputs → Component Placement → CSV → includes: Reference, Value, Footprint, X, Y, Rotation, Side		
BOM export	File → Fabrication Outputs → BOM → use kibom plugin or interactive HTML BOM for richer output		
JLC rotation fix	Many JLC footprints have 90° or 180° offset – use 'jlcpcb-kicad-footprints' library or correct rotation in centroid CSV		
ORDERING CHECKLIST			
Board dimensions	Verify board size in mm in fab order form	Layer count	Match ordered layers to your design
Surface finish	HASL-Pb free (cheapest) / ENIG (for fine-pitch)	Solder mask	Green/Red/Blue/Black – green cheapest and fastest
Silkscreen	White on green/black; black on white	Controlled imp.	Request only if design needs it (adds cost)
Quantity & stencil	Order stencil with first prototype run	Fab DFM check	Upload Gerbers to JLC/PCBWay DFM tool before confirming
NOTE: Always order a 3D print or laser-cut mechanical mockup before ordering PCBs for enclosure-constrained designs. Verify connector positions and mounting hole positions match mechanical drawings.			

21. PCB DESIGN GOLDEN RULES & BEST PRACTICES

THE GOLDEN RULES OF PCB DESIGN	
SCHEMATIC RULES	
Rule S1	Schematic is the engineering document. If it is wrong, the PCB is wrong. NEVER modify netlist manually to fix schematic errors.
Rule S2	Name EVERY signal net. Unnamed nets (Net-(R3-Pad1)) are impossible to debug. Name at minimum: clocks, power rails, data buses, interrupt lines.
Rule S3	Run ERC and fix ALL errors before opening PCB editor. Warnings must be reviewed and documented – never silently ignored.
Rule S4	Add decoupling caps in schematic, not as an afterthought. Every IC VCC pin needs one. Include in BOM and assign footprint before layout.
LAYOUT RULES	
Rule L1	Current needs a return path. Every signal current comes from somewhere and returns to source. Ensure that path is short and has low impedance.
Rule L2	Never split a ground plane under a signal trace. The return current MUST have a direct path – splits force long detours and create EMI loops.
Rule L3	Decoupling capacitors: the shorter the path from cap to VCC pin, the better. Via on GND side of cap goes DIRECTLY to GND plane, not a long trace.
Rule L4	Switching regulator layout: make the hot loop (Vin cap → high-side FET → low-side FET → input cap GND) as small as physically possible.
Rule L5	Run DRC and fix ALL errors before generating Gerbers. DRC warnings must be reviewed. A board with DRC errors will almost certainly have a problem.
HIGH-SPEED RULES	
Rule H1	When trace length > $Tr/(2 \times Tpd)$, the trace is a transmission line. Apply termination. On FR4: traces > 25mm at 1Gbps need impedance control.
Rule H2	Differential pairs: route as a pair from source to load. Never split them. Match intra-pair length to within 5mil. Maintain coupling (close spacing) for full benefit.
Rule H3	Crystal: place as close as possible to IC pins. GND guard ring on all sides. No other traces through crystal area. Dedicated GND pour, stitched to main GND.
PROCESS RULES	
Rule P1	Use version control (Git) for PCB designs. Commit at every milestone: after schematic, after placement, after routing, before Gerber export.
Rule P2	Document design decisions. Add notes to schematic (text blocks) explaining why: why this value, why this topology, why this layout choice.
Rule P3	Review with fresh eyes: wait 24h, print the schematic and review on paper, have a colleague review. Errors are invisible when you created them.
Rule P4	Prototype before production: order 5 prototype boards, assemble and test one. Fix all issues before ordering production quantity.
NOTE: The most expensive PCB mistake is the one that passes your review. The second most expensive is the one that doesn't get reviewed. Always review. Always test. Always document.	

22. PCB TESTING, BRING-UP & DEBUGGING

BOARD BRING-UP SEQUENCE			
BARE BOARD TESTS (before assembly)			
Visual inspection	Check for physical damage, delamination, mask voids	Continuity test	Every net: no opens, no shorts between different nets
Impedance test	TDR or VNA on controlled impedance traces + coupon	Hi-pot test	High-voltage isolation test for mains-connected boards

BRING-UP SEQUENCE (after assembly)			
1. Current limit	Set bench PSU to expected I_{max} + 20mA – watch for smoke	2. Power rails	Check VCC, VCORE, VDDA with DMM before powering MCU
3. Clock verify	Probe crystal / oscillator output with oscilloscope	4. Reset release	Monitor RESET line – verify it deasserts correctly
5. Comms test	UART loopback → SPI flash ID → I2C scan → CAN/ETH	6. Peripherals	GPIO toggle → ADC reading → PWM output → ISR
7. Load testing	Full operational load – measure power, check temps	8. Stress test	Thermal cycling, vibration, EMC pre-compliance test
DEBUG TOOLS & TECHNIQUES			
Oscilloscope probing	Use X10 probe; ground clip as short as possible to TP	Current probe	Hall-effect clamp for supply current measurement
Logic analyser	Saleae Logic / PulseView – 8/16ch digital capture	Spectrum analyser	EMI pre-compliance: near-field probe + SA (RSA306B)
Thermal camera	FLIR/Seek – identify hot spots during load test	De-solder station	Hot air + tweezers – swap suspect components safely
Oscilloscope power	Measure PDN: scope probe across VCC/GND cap during load	Ripple target	< 10mV for digital logic; < 1mV for precision analogue
COMMON ASSEMBLY FAULTS & FIXES			
Solder bridge	Flux + solder wick – wick from pad, not wire	Cold joint	Reflow with hot air + flux – glossy finish = good solder
Tomb-stoning	0402/0603 standing on end – reflow profile / pad mismatch	Missing paste	Reflow with paste manually applied + hot air
Part polarity	Check cap (stripe = negative), LED, diode orientation	Wrong part	Remove with hot air, verify MPN, replace and reflow
NOTE: Keep a bring-up log: date, what you tested, result, scope screenshots. This is invaluable when debugging issues weeks later or when someone else inherits the board.			

23. SPECIAL PCB TYPES — FLEX, RIGID-FLEX, HDI

FLEX PCB DESIGN RULES			
FLEX PCB MATERIALS & MINIMUMS (IPC-2223)			
Base material	Polyimide (Kapton): 12.5µm-125µm thick	Copper	0.5oz (17µm) for dynamic flex; 1oz for static flex
Min trace width	0.1mm (dynamic flex); 0.075mm (static)	Min clearance	0.1mm minimum trace-to-trace
Bend radius	Dynamic: 10× total thickness minimum	Static bend	6× total thickness minimum
No vias in bend	Keep vias and pads ≥ 1mm from bend zone	Teardrop pads	Add teardrops at trace-to-pad transitions – reduces stress
Trace orientation	Route traces perpendicular to bend axis in bend zone	Avoid crossing	No trace crossings in bend area – use hatched fill
RIGID-FLEX DESIGN (IPC-2223C)			
Stiffener	FR4 or polyimide stiffener in connector area for strength	Coverlay	Adhesive polyimide overlay – replaces solder mask on flex
Via-in-pad	Plated flex vias must be encapsulated – no bare vias	Layer transition	Stepped board edge where rigid meets flex
ZIF connector	ZIF (Zero Insertion Force) on flex tail – spec insertion cycles	Tail thickness	Match ZIF connector specification exactly
HDI (HIGH DENSITY INTERCONNECT)			
Micro-vias	0.1mm laser drill – only connects adjacent layers	Via stacking	Stacked micro-vias (costly) vs staggered (cheaper)
Capture pad	0.2mm pad for 0.1mm micro-via – very tight	ELIC	Every Layer Interconnect – micro-vias on every layer
BGA escape	Use micro-vias for fine-pitch BGA (< 0.8mm pitch) escape	Dog-bone	Via behind pad with trace to via – for 1mm+ BGA pitch
ALUMINIUM / METAL CORE PCB (MCPCB)			
Use case	LED lighting, high-power modules – better thermal	Thermal cond.	1-3 W/m·K for MCPCB vs 0.3 W/m·K for FR4
Dielectric layer	Thin dielectric between copper and Al – 50-150µm	Single layer	Usually 1-layer copper on top of Al substrate
Design rules	Min 0.2mm clearance; no plated holes to Al core	Cutouts	Punched, not drilled – check with manufacturer
NOTE: Flex PCB design requires flex-experienced fabricator. Always confirm bend radius, dynamic vs static flex, and stiffener requirements with your fab BEFORE starting layout. Mistakes in flex design are expensive.			

24. PCB COST REDUCTION STRATEGIES

PCB FABRICATION COST REDUCTION			
BOARD DESIGN FOR COST			
Fewer layers	2-layer < 4-layer < 6-layer – halve layers where possible	Standard size	JLC cheapest: 100×100mm or smaller; use full panel
Standard trace	Avoid 0.09mm traces – use 0.127mm+ to stay in std process	Standard via	0.3mm drill / 0.6mm pad – avoid micro-via premium
HASL finish	Cheaper than ENIG – use only if fine-pitch parts absent	2-layer for GND	Pour GND on bottom layer – avoids 4-layer premium
Board panelisation	Fit multiple designs on one panel – amortise setup cost	V-score panel	JLC panels: max 500×500mm – fit as many as possible
ASSEMBLY COST REDUCTION			

Single-side SMD	All SMD on top – one reflow pass, cheaper PCBA	Avoid BGA	BGAs require X-ray inspection – adds assembly cost
Standard pitch	≥ 0.65mm pitch parts: cheaper to assemble than 0.4mm	Larger packages	SOIC > TSSOP > QFN for assembly cost, ease
JLCPCB Basic parts	Use 'Basic' parts from LCSC – no extended part handling fee	Extended parts	+\$3/type for non-basic parts – minimise unique extended
Combine BOM	Same value/footprint = same part – reduces part types	Standard values	E12 or E24 series values – always in stock
COMPONENT COST REDUCTION			
LCSC / JLCPCB	Chinese marketplace – fraction of Mouser/Digi-Key price	JLCPCB SMT	Best value for small volumes < 100 boards
Octopart search	Price comparison across all distributors	EOL avoidance	Check last-buy date + stock level during design
Alternative MPN	List 2 alternative MPNs per critical component	DNP parts	Design Now, Place Never – include in BOM, skip in order
NOTE: Cheapest design = fewest layers, all-top-side SMD, 0402/0603 standard packages, LCSC basic parts, board size ≤ 100×100mm. Cost optimise AFTER design is functionally correct, not before.			

25. QUICK REFERENCE — UNITS, STANDARDS & CONSTANTS

PCB UNITS & CONVERSION			
1 mil (thou)	0.0254 mm (1/1000 inch)	1 mm	39.37 mil
1 inch	25.4 mm = 1000 mil	IU (internal)	KiCAD: 1 IU = 1 nm (10 ⁻⁶ mm)
pcbnew.FromMM(x)	Converts mm to KiCAD IU	pcbnew.ToMM(x)	Converts KiCAD IU to mm
STANDARD TRACE WIDTHS (reference)			
Signal (default)	0.15–0.2mm (6–8mil)	Power (1A)	0.4mm (16mil) minimum
Power (2A)	0.8mm (31mil)	Power (5A)	2.0mm (79mil)
USB D+/D- (90Ω)	0.16mm – check fab stackup	50Ω microstrip	Depends on stackup – use calculator
KEY IPC STANDARDS REFERENCE			
IPC-2221B	Generic PCB design standard	IPC-2222	Rigid organic PCB design
IPC-2223C	Flexible/Rigid-flex design	IPC-7351C	Land pattern and footprint standard
IPC-A-610	Acceptability of electronic assemblies	IPC-6012F	Qualification of rigid PCBs
IPC-2141C	Controlled impedance design	J-STD-001	Soldering requirements
MATERIAL PROPERTIES (FR4)			
Dielectric constant	ε _r = 4.3–4.7 @ 1GHz (varies with frequency)	Loss tangent	tanδ = 0.018–0.020 @ 1GHz
T _g (glass transition)	T _g = 130°C (standard); 170°C (high-T _g)	T _d (decomp.)	T _d > 300°C (FR4)
Thermal cond.	0.3 W/m·K (poor) – use MCPCB or vias for heat spreading	CTE	17 ppm/°C in-plane; 55 ppm/°C Z-axis
ELECTROMAGNETIC CONSTANTS (PCB)			
Speed in FR4	v = c/√ε _r ≈ 1.45×10 ⁸ m/s ≈ 0.48c	Delay per mm	≈ 6.9 ps/mm on FR4 (ε _r =4.4)
λ at 1GHz in FR4	λ = v/f ≈ 145mm (λ/10 = 14.5mm)	λ at 2.4GHz	λ ≈ 60mm in FR4 (λ/10 = 6mm antenna keepout)
Skin depth in Cu	δ = 1/√(πfμσ) – 2.1μm at 1GHz; 0.66μm at 10GHz	1oz Cu @ 1GHz	35μm >> skin depth → full 1oz participates at GHz
NOTE: For SA fabrication: PCBWay, JLCPCB, and Elmatica all ship to South Africa. JLCPCB offers DDP (Delivered Duty Paid) — no import surprises. Factor 2–3 week lead time + shipping for prototypes.			

26. STM32H755ZIT6 — POWER WIRING, DECOUPLING & PROGRAMMING

STM32H755ZIT6 — LQFP144 OVERVIEW			
Package	LQFP144 — 144-pin, 0.5mm pitch, 20×20mm body	Cores	Cortex-M7 @ 480MHz + Cortex-M4 @ 240MHz
Flash	2MB dual-bank (Bank1 + Bank2, 1MB each)	RAM	1MB SRAM: DTCM(128K)+ITCM(64K)+AXI(512K)+D2/D3
Supply range	VDD: 1.71–3.6V VDDA: 1.62–3.6V	Package marking	STM32H755ZIT6 = LQFP144, Temp -40°C to +85°C (I grade)
VDD POWER PINS (connect all to +3.3V with 100nF per pin)			
VBAT — Pin 8	Backup domain supply: RTC, backup SRAM, LSE osc.	Recommended	Connect to 3.3V with 100nF. Add coin cell via diode if backup needed
VDD — Pin 7	Digital supply (I/O Bank A, partial)	VDD — Pin 13	Digital supply (I/O Bank A/B)
VDD — Pin 19	Digital supply (I/O Bank B)	VDD — Pin 32	Digital supply (I/O Bank C)
VDD — Pin 42	Digital supply (I/O Bank C)	VDD — Pin 56	Digital supply (I/O Bank D)
VDD — Pin 71	Digital supply (I/O Bank E)	VDD — Pin 79	Digital supply (I/O Bank E/F)
VDD — Pin 92	Digital supply (I/O Bank F/G)	VDD — Pin 106	Digital supply (I/O Bank G/H, SWD region)
VDD — Pin 119	Digital supply (I/O Bank H/I)	VDD — Pin 129	Digital supply (I/O Bank I/J)
VDD — Pin 144	Digital supply (I/O Bank J/K)	Total VDD pins	13× VDD pins — each MUST have 100nF cap to GND
USB & ANALOGUE SUPPLY PINS			
VDD33_USB — Pin 91	USB 3.3V supply — connects to VDD (+3.3V) rail	Decoupling	1µF + 100nF directly at pin
VDD50_USB — Pin 90	USB 5V supply — connects to +5V_USB (VBUS/D+ ref)	Decoupling	1µF + 100nF. Needed even if not using USB
VDDA — Pin 36	Analogue supply for ADC, DAC, PLL, RNG, temperature sensor	Decoupling	1µF tantalum/electrolytic + 10nF ceramic. Ferrite bead from VDD to VDDA (600R@100MHz)
INTERNAL VOLTAGE REGULATOR PINS (VDDCORE system)			
VDDSMPS — Pin 16	Input to internal SMPS regulator — connects to +3.3V	Decoupling	4.7µF + 100nF at pin. SMPS output feeds VDDCORE (1.2V)
VDDLDO — Pin 70	Input to internal LDO — connects to VDDCORE output	VDDLDO — Pin 105	Same: input to internal LDO, connect to VDDCORE rail
VDDLDO — Pin 143	Input to internal LDO — connects to VDDCORE	VDDCORE rail	~1.2V generated internally — route to VDDLDO pins only
NOTE: VDDSMPS generates VDDCORE (1.2V internal). Route VDDCORE net (from SMPS output) ONLY to VDDLDO pins. Never connect VDDLDO to 3.3V. The SMPS output is NOT exposed as a PCB pin — it is internal to the IC. Use BYPASS mode if SMPS is disabled.			
VCAP PINS — INTERNAL LDO OUTPUT BYPASS			
VCAP — Pin 68	Internal core voltage regulator output — DO NOT connect to external supply	Decoupling	2.2µF ceramic (X5R/X7R) to GND — exactly as shown in Image 2
VCAP — Pin 103	Internal core voltage regulator output — bypass cap to GND	Decoupling	2.2µF ceramic to GND

VCAP — Pin 140	Internal core voltage regulator output – bypass cap to GND	VCAP rule	Each VCAP pin MUST have its own 2.2µF cap. Never share. Never connect together via trace.
NOTE: VCAP is the output of the internal LDO/SMPS core regulator (~1.2V). It MUST have 2.2µF ±20% (X5R or better) per pin. DO NOT connect VCAP pins together. DO NOT load VCAP with external circuitry. ST AN4879 is the reference application note for H7 power supply.			
VSS / GND PINS — ALL MUST CONNECT TO GND			
VSS — Pin 6	Digital ground return – primary VSS	VSS — Pin 11	Digital ground (also referenced as VSS)
VSS — Pin 17	Digital ground	VSS — Pin 27	Digital ground
VSS — Pin 38	Digital ground	VSS — Pin 50	Digital ground
VSS — Pin 65	Digital ground	VSS — Pin 76	Digital ground
VSS — Pin 89	Digital ground	VSS — Pin 100	Digital ground
VSS — Pin 116	Digital ground	VSS — Pin 127	Digital ground
VSS — Pin 138	Digital ground	VSSA — Pin 34	Analogue ground – star-connect to GND at single point
VSSMPS — Pin 14	SMPS ground – connect to GND (same as VSS)	GND rule	All VSS, VSSA, VSSMPS connect to the same GND plane
NOTE: VSSA (analogue ground) must connect to the main GND plane but ideally at a single star point away from noisy digital switching. Route VDDA supply through ferrite bead (600Ω@100MHz) from VDD. Route VSSA back to GND via a 0Ω resistor or ferrite bead for the star connection.			

STM32H755ZIT6 — COMPLETE DECOUPLING TABLE**DECOUPLING CAPACITOR PLACEMENT RULES**

Rule 1 — VDD	100nF X7R 0402 on EVERY VDD pin (13 caps total). Place on same side as IC, as close to pin as physically possible. GND via goes directly to GND plane.
Rule 2 — Bulk	10µF X5R 0603 per power domain: one near the VDD entry to the MCU section. This handles lower-frequency transients that 100nF cannot.
Rule 3 — VDDA	10nF + 1µF (tantalum or X5R electrolytic) + ferrite bead (600Ω@100MHz, 500mA) in series from VDD to VDDA. VSSA connect with 0Ω resistor to GND.
Rule 4 — VCAP	2.2µF ±20% X5R/X7R 0402 per VCAP pin (3 caps total: pins 68, 103, 140). Each cap isolated – no shared traces between VCAP pins.
Rule 5 — VBAT	100nF + 1µF on VBAT pin 8. If using backup battery: Schottky diode from VDD → VBAT anode, battery to VBAT directly (diode OR topology).
Rule 6 — VDDSMPS	4.7µF + 100nF on VDDSMPS (pin 16). Add 100nF on each VDDLDO pin (70, 105, 143). SMPS output (VDDCORE) needs 4.7µF near the SMPS inductor (internal – but near the IC).

POWER SUPPLY ARCHITECTURE DECISION

SMPS mode	Recommended: lower power, VDDCORE generated by SMPS	LDO mode	Simpler: bypass SMPS, LDO provides VDDCORE. More heat.
BYPASS mode	External VDDCORE: supply 1.2V externally to VDDLDO pins	PWR_CR3 reg	Configure SMPS/LDO/BYPASS in firmware: PWR_CR3.SMPSEN
SMPS external L	SMPS mode requires 2.2µH inductor between VDDSMPS and VDDCORE (external). Use Murata LQH series or equivalent low-DCR. Current rating ≥ 500mA.		
SMPS output cap	VDDCORE: 10µF + 1µF ceramic at SMPS inductor output. Route VDDCORE only to VDDLDO pins – not to anything else.		

NOTE: See ST datasheet DS12930 and application note AN4938 (STM32H7 power supply) and AN4879 (hardware guidelines). Use ST's STM32CubeH7 power_supply example project to configure PWR_CR3 for your chosen supply mode.

STM32H755ZIT6 — PROGRAMMING & DEBUG PINS (LQFP144)**SWD (SERIAL WIRE DEBUG) — PRIMARY DEBUG INTERFACE (2 pins)**

SWDIO — PA13 / Pin 105	Serial Wire Data I/O – bidirectional data line	Pull	Internal pull-up (50kΩ). External 10kΩ pull-up to VDD recommended
SWDCLK — PA14 / Pin 106	Serial Wire Clock – from programmer to MCU	Pull	Internal pull-down. External 10kΩ pull-down to GND recommended

NRST — Pin 25	Active-low reset input. 100nF cap to GND, 10kΩ to VDD	SWD connector	10-pin Cortex Debug (Tag-Connect TC2030) or 0.1in header
JTAG — FULL DEBUG INTERFACE (5 pins, subset of SWD)			
JTMS/SWDIO — PA13/105	Test Mode Select (JTAG) / Data (SWD)	JTCK/SWDCLK — PA14/106	Test Clock
JTDI — PA15 / Pin 107	Test Data In — JTAG only (not needed for SWD)	JTDO/SWO — PB3/133	Test Data Out / Trace SWO (printf without UART)
NJTRST — PB4 / Pin 134	JTAG TAP reset — pull up to VDD if not used	SWO / TRACESWO	PB3: connect to programmer SWO pin for ITM printf output
BOOT CONFIGURATION PINS			
BOOT0 — Pin 60	Boot mode selection: 0=Flash, 1=System memory (bootloader)	Pull-down	10kΩ pull-down to GND for normal boot from Flash
BOOT pin control	In STM32H7: BOOT0 only + BOOT_ADD0/1 option bytes	Bootloader mode	BOOT0 HIGH at reset → enter ROM bootloader (UART/USB DFU/SPI)
BOOT0 circuit	BOOT0(pin60)—10kΩ—GND AND BOOT0—Jumper/switch—VDD (for field bootloader access). Alternatively use dedicated BOOT0 pad.		
STM32H7 DEBUG CONNECTOR STANDARDS			
10-pin Cortex Debug	2×5, 1.27mm pitch — standard ARM Cortex SWD	Pins	1:VCC_Target 2:SWDIO 3:GND 4:SWDCLK 5:GND 6:SWO 7:NC 8:NC 9:GND 10:nRESET
20-pin JTAG	2×10, 2.54mm pitch — full JTAG + SWD	Tag-Connect	TC2030-CTX-NL: no PCB connector needed — uses spring-pin clip
4-pin SWD minimal	VCC, SWDIO, SWDCLK, GND — absolute minimum for debug	TC2030-IDC-NL	Tag-Connect 6-pin: SWD + nRESET + SWO — recommended for production boards
PROGRAMMING INTERFACE PULL RESISTORS (always include on PCB)			
SWDIO pull-up	10kΩ from SWDIO (PA13/Pin105) to VDD — prevents floating when no programmer attached. DNP if debug only during development.		
SWDCLK pull-dn	10kΩ from SWDCLK (PA14/Pin106) to GND — holds clock low when no programmer attached.		
NRST filter	100nF from NRST (Pin25) to GND — filters ESD/noise on reset line. 10kΩ pull-up to VDD already internal but add external too.		
SWO optional	Connect PB3 (Pin133) to programmer SWO pin via 33Ω series resistor. Enables ITM/SWO trace output (printf-style debug without UART).		
NOTE: SWD (SWDIO + SWDCLK + GND + VCC) is the minimum viable debug interface. Always include a Tag-Connect TC2030-IDC-NL landing on the PCB — it adds zero cost (6 pads + 3 mounting holes) and eliminates need for a connector in production.			

RECOMMENDED PROGRAMMERS — STM32H755ZIT6

DEVELOPMENT PROGRAMMERS / DEBUGGERS			
ST-LINK/V3 MINIE	ST official: SWD+JTAG+SWO, USB-C, UART bridge. Best value.	Cost	~\$12 USD — highly recommended for all development work
ST-LINK/V3E	Enhanced: SWD+JTAG+SWO+UART, 1.65-5V VCC auto-detect	Cost	On STM32H7 Nucleo boards (detachable) — free with eval board
ST-LINK/V3 SET	Full kit: STLINK/V3 + STDC14 cable + adapters	Protocol	SWD 50MHz, JTAG 33MHz, SWO 24MHz, VCP UART 16Mbps
J-Link BASE	SEGGER: gold standard, fastest, all protocols	Cost	~\$500 USD (full), ~\$18 (EDU — for non-commercial only)
J-Link EDU Mini	SEGGER educational licence: SWD+JTAG+SWO, limited commercial	Speed	Up to 50MHz SWD. Best for open-source/student projects
CMSIS-DAP / DAPLink	Open standard: supported by PlatformIO, pyOCD, OpenOCD	Cost	< \$5 USD clones available. Slower but universal
BlackMagicProbe v2.3	Open-source JTAG/SWD probe with built-in GDB server	Benefit	No OpenOCD needed — GDB connects directly to probe

Nucleo-H755ZI-Q	STM32H755 eval board: has ST-LINK/V3E built-in detachable	Benefit	Powers and programs your target via SWD cable – ideal for first bring-up
MASS PRODUCTION PROGRAMMERS			
ST-LINK/V3 MODS	ST module for integration into custom test jigs (SWD)	Interface	SWD + UART. Can be automated via STM32CubeProgrammer CLI
STLINK-V3PWR	ST-LINK with power measurement: SWD+power monitoring	Use case	Production test + current consumption verification in one tool
Segger J-Link PRO	High-speed production: 25MHz SWD, multi-target, licensing	Automation	Command-line: JLink.exe - CommandFile flash_cmd.txt (scriptable)
Segger FlashPro	Gang programmer: programs up to 32 targets simultaneously	Use case	High-volume production line (>10K units)
ISSI Cyclone FX	P&E Micro gang programmer: standalone, no PC required	Storage	Stores encrypted image locally – operator just presses button
STM32CubeProgrammer	ST official GUI+CLI: SWD/JTAG/UART/USB DFU/SPI bootloader	CLI syntax	STM32_Programmer_CLI -c port=SWD -d firmware.hex 0x08000000 -rst
UART Bootloader DFU	STM32H7 ROM bootloader: USART1/3, USB, SPI, I2C, CAN	BOOT0 high	Set BOOT0=1, reset MCU → enters factory ROM bootloader automatically
USB DFU mode	USB Full-Speed DFU: PA11/PA12 USB FS pins – no UART needed	Tool	dfu-util (open source) or STM32CubeProgrammer USB DFU mode
PROGRAMMER COMPARISON SUMMARY			
Dev + Debug	ST-LINK/V3 MINIE (best value) or J-Link EDU Mini (fastest)	Bring-up	Use Nucleo board as debugger + power source for first test
Low volume prod.	ST-LINK/V3 MODS + STM32CubeProgrammer CLI in test jig	High volume	Segger FlashPro gang programmer or ISSI Cyclone FX standalone
Field firmware update	UART DFU (bootloader) or USB DFU – no JTAG connector needed	Security	Enable RDP Level 1 after production flash – prevents readback
NOTE: For SA production: ST-LINK/V3 MINIE is the best value development programmer (available from Micro Robotics / RS Components SA). For automated production flashing, use STM32CubeProgrammer CLI in a Python/batch script triggered by test jig GPIO. Enable read protection (RDP1) after programming to protect firmware IP.			

STM32H755ZIT6 — KiCAD Schematic Wiring Checklist (per screenshots)

// — VDD PINS (Image 1) – all connect to +3.3V net —	
// Pin 8 VBAT	→ +3.3V (100nF to GND, add 1µF if using RTC/backup)
// Pin 7 VDD	→ +3.3V (100nF to GND)
// Pin 13 VDD	→ +3.3V (100nF to GND)
// Pin 19 VDD	→ +3.3V (100nF to GND)
// Pin 32 VDD	→ +3.3V (100nF to GND)
// Pin 42 VDD	→ +3.3V (100nF to GND)
// Pin 56 VDD	→ +3.3V (100nF to GND)
// Pin 71 VDD	→ +3.3V (100nF to GND)
// Pin 79 VDD	→ +3.3V (100nF to GND)
// Pin 92 VDD	→ +3.3V (100nF to GND)
// Pin 106 VDD	→ +3.3V (100nF to GND) ← NOTE: same pin group as SWDCLK
// Pin 119 VDD	→ +3.3V (100nF to GND)
// Pin 129 VDD	→ +3.3V (100nF to GND)
// Pin 144 VDD	→ +3.3V (100nF to GND)
// — USB SUPPLY PINS —	
// Pin 91 VDD33_USB	→ +3.3V (1µF + 100nF to GND)
// Pin 90 VDD50_USB	→ +5V_USB (1µF + 100nF to GND) ← needs 5V even if USB unused
// — ANALOGUE SUPPLY —	
// Pin 36 VDDA	→ +3.3V via 600R@100MHz ferrite bead (1µF + 10nF to GND)
// Pin 34 VSSA	→ GND via 0Ω resistor (star point connection)


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// — INTERNAL REGULATOR PINS —
// Pin 16 VDDSMPS → +3.3V (4.7µF + 100nF to GND)
//                                     External 2.2µH inductor: VDDSMPS → L1 → VDDCORE net
// Pin 70 VDDLDO → VDDCORE net (100nF to GND)
// Pin 105 VDDLDO → VDDCORE net (100nF to GND)
// Pin 143 VDDLDO → VDDCORE net (100nF to GND)
// VDDCORE → 10µF + 1µF to GND (near inductor output)

// — VCAP PINS (Image 2) — internal LDO output bypass —
// Pin 68 VCAP → 2.2µF X5R 0402 to GND (DO NOT connect to supply!)
// Pin 103 VCAP → 2.2µF X5R 0402 to GND (separate cap, no shared net)
// Pin 140 VCAP → 2.2µF X5R 0402 to GND (separate cap, no shared net)

// — VSS PINS (Image 3) — all connect to GND —
// Pin 6 VSS → GND
// Pin 11 VSS → GND
// Pin 14 VSSMPS → GND (SMPS return — same GND plane)
// Pin 17 VSS → GND
// Pin 27 VSS → GND
// Pin 34 VSSA → GND (via 0Ω/ferrite star connection — separate from DGND trace)
// Pin 38 VSS → GND
// Pin 50 VSS → GND
// Pin 65 VSS → GND
// Pin 76 VSS → GND
// Pin 89 VSS → GND
// Pin 100 VSS → GND
// Pin 116 VSS → GND
// Pin 127 VSS → GND
// Pin 138 VSS → GND

// — PROGRAMMING / DEBUG PINS —
// Pin 105 PA13/SWDIO → 10-pin debug header pin 2 (10kΩ pull-up to VDD)
// Pin 106 PA14/SWDCLK → 10-pin debug header pin 4 (10kΩ pull-down to GND)
// Pin 25 NRST → 10-pin debug header pin 10 (100nF to GND, 10kΩ to VDD)
// Pin 133 PB3/SWO → 10-pin debug header pin 6 (33Ω series) — optional SWO trace
// Pin 60 BOOT0 → 10kΩ to GND + solder jumper/test point to VDD

// — 10-PIN CORTEX DEBUG HEADER (1.27mm pitch, 2×5) —
// Pin 1: VCC_TARGET (connect to 3.3V via 10Ω series resistor)
// Pin 2: SWDIO (PA13) Pin 3: GND
// Pin 4: SWDCLK (PA14) Pin 5: GND
// Pin 6: SWO (PB3) Pin 7: KEY (no pin)
// Pin 8: NC Pin 9: GND
// Pin 10: nRESET (NRST)
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STM32H755ZIT6 — POWER SUPPLY SCHEMATIC BLOCKS	
BLOCK 1 — 3.3V DIGITAL SUPPLY TO MCU (main VDD)	
Supply source	3.3V LDO or buck converter output → bulk cap (10µF) → ferrite bead (optional) → 3.3V rail → fan out to all VDD pins
Per VDD cap	Each VDD pin: 100nF X7R 0402 placed within 0.5mm of pin, GND via straight down to GND plane
KiCAD net name	Use net label '+3.3V' on all VDD pins — matches Image 1. One power symbol per sub-sheet is cleaner.
BLOCK 2 — ANALOGUE SUPPLY (VDDA)	
VDDA filter	+3.3V —[FB1: 600Ω@100MHz / 500mA]— VDDA node —[1µF + 10nF to VSSA]
VSSA connection	VSSA —[R_STAR: 0Ω]— GND (place at star ground point, far from digital switching)
KiCAD net name	Use 'VDDA' and 'VSSA' as separate net names in KiCAD — assign VSSA to analogue GND power symbol
BLOCK 3 — INTERNAL SMPS (VDDCORE)	
SMPS circuit	+3.3V —[4.7µF to GND]— VDDSMPS(Pin16) VDDSMPS —[L1: 2.2µH]— VDDCORE —[10µF + 1µF to GND]
VDDCORE to LDO	VDDCORE —[100nF each]— VDDLDO(Pin70) VDDLDO(Pin105) VDDLDO(Pin143)

KiCAD net name	Use 'VDDCORE' net label on VDDCORE rail. This net only connects to: L1 output, VDDLDO pins, and their bypass caps.
BLOCK 4 — USB SUPPLY	
USB 3.3V	+3.3V —[1 μ F + 100nF to GND]— VDD33_USB(Pin91) — always powered from 3.3V rail
USB 5V	+5V_USB(VBUS) —[1 μ F + 100nF to GND]— VDD50_USB(Pin90) — required even if only USB FS is used
USB protection	VBUS —[F1: 500mA PTC resettable fuse]—[D_TVS: USBLC6-2SC6 or PRTR5V0U2X]— USB connector D+/D-
BLOCK 5 — BACKUP SUPPLY (VBAT)	
Simple VBAT	+3.3V —[100nF + 1 μ F to GND]— VBAT(Pin8) — sufficient if no battery backup needed
With battery	+3.3V —[D1: BAT54C Schottky]— CR2032 —[D2: BAT54C Schottky]— → VBAT(Pin8) —[1 μ F to GND]
VBAT function	Maintains RTC, backup SRAM (4KB), LSE oscillator, BOR on VBAT domain even when VDD is off
NOTE: Reference: ST DS12930 (STM32H755 datasheet), AN4938 (STM32H7 power supply), AN4879 (STM32H7 hardware dev guidelines), RM0399 (STM32H745/755 reference manual). Always verify pin numbers against the SPECIFIC package variant datasheet.	

PCB DESIGN QUICK REFERENCE

DESIGN CHECKLIST — BEFORE SENDING TO FAB			
ERC passed	Zero errors in Electrical Rules Check	DRC passed	Zero errors in Design Rules Check
Footprints verified	Every footprint checked against datasheet	3D check	3D viewer shows no collisions or missing parts
Silkscreen clean	No text under pads; all ref designators visible	Courtyard clean	No overlapping courtyards
Gerbers verified	Inspected in Gerber viewer — all layers correct	Drill check	Drill map shows all holes in correct positions
BOM complete	Every component has MPN, value, footprint, quantity	Assembly files	Centroid CSV exported and verified
Stackup confirmed	Layer stackup matches fab's available options	Impedance	Controlled impedance traces confirmed with fab calculator
Design rule file	Loaded fab's .kicad_dru file and DRC clean	Version control	Design committed to Git with meaningful commit message

PCB Design Cheat Sheet • 26 Sections • Workflow | Schematic | Footprints | Stackup | Placement | Routing | SI | PDN | EMC | DRC/DFM | KiCAD | Shortcuts | Python | Libraries | High-Speed | RF | Thermal | Testing | Flex | Cost